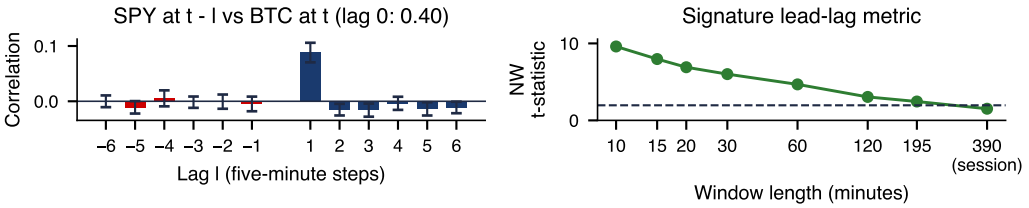
SPY at t - l vs BTC at t (lag 0: 0.40)
Correlation
0.1
0.0
−6 −5 −4 −3 −2 −1 1 2 3 4 5 6
Lag l (five-minute steps)
Signature lead-lag metric
NW t-statistic
10
0
10 15 20 30 60 120 195 390 (session)
Window length (minutes)
Lag l > 0: SPY moves first
Lag l < 0: Bitcoin moves first
Pointwise 95% interval (Newey-West)
t = 1.96 (pointwise)
Mean daily Lévy area (SPY, BTC)